

HOKMARKSBERGET, Sweden

WMO: 022920

Lat: 64.4358N

Lon: 21.2800E

Elev: 92

StdP: 100.22

Time Zone: 1.00 (EUC)

Period: 01-16

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.1	-18.5	-23.7	0.4	-19.4	-21.1	0.6	-17.4	7.4	-2.5	6.8	-2.6	1.6	0	0.612

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.0	25.0	16.3	23.1	15.2	21.4	14.6	18.2	21.9	17.1	20.4	16.2	19.3	2.7	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.9	12.2	18.8	16.0	11.5	17.8	15.0	10.8	17.1	51.8	22.2	48.3	20.4	45.8	19.3	22.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
6.6	5.9	5.2	-23.1	27.7	3.0	2.0	-25.2	29.2	-27.0	30.4	-28.7	31.5	-30.9	33.0		
			DB	WB	DB	WB	DB	WB	DB	WB	DB	WB	DB	WB		
			-22.5	19.0	3.4	1.4	-25.0	20.0	-26.9	20.8	-28.9	21.6	-31.3	22.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	4.2	-6.7	-6.7	-1.9	2.6	8.0	13.2	16.5	15.2	10.2	4.0	-0.8	-4.3
	DBStd	9.12	5.73	6.24	4.37	3.42	3.67	3.14	2.83	2.99	3.26	3.84	4.30	5.59
	HDD10.0	2662	518	468	369	223	83	6	0	2	37	187	324	444
	HDD18.3	5191	777	702	627	472	319	157	71	103	243	444	574	703
	CDD10.0	538	0	0	0	1	23	103	203	163	44	1	0	0
	CDD18.3	25	0	0	0	0	0	3	15	7	0	0	0	0
	CDH23.3	156	0	0	0	0	2	31	92	32	0	0	0	0
	CDH26.7	23	0	0	0	0	0	4	17	2	0	0	0	0
Wind	WSAvg	2.4	2.2	2.6	2.5	2.5	2.6	2.4	2.4	2.1	2.2	2.3	2.4	2.3
Precipitation	PrecAvg	637	45	35	38	34	39	45	60	76	72	70	71	52
	PrecMax	900	101	83	115	126	101	147	164	176	161	185	138	170
	PrecMin	396	4	3	1	1	6	4	10	15	14	18	13	3
	PrecStd	104	23	22	20	25	24	28	37	45	32	36	28	29
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.0	6.6	10.7	16.4	22.6	26.4	28.6	26.5	20.2	13.9	9.3	5.8
		MCWB	3.2	3.2	4.7	8.7	12.4	14.9	18.4	18.0	13.6	9.3	7.5	2.4
	2%	DB	3.4	4.5	7.9	12.7	19.3	23.4	25.8	23.9	17.8	11.6	7.3	4.5
		MCWB	1.5	1.7	3.2	6.5	11.3	14.2	16.8	15.8	12.6	9.0	6.2	3.0
	5%	DB	1.7	2.2	6.0	10.3	16.7	21.3	23.6	22.1	16.4	10.3	5.9	3.3
		MCWB	1.0	0.1	2.3	4.5	9.3	13.0	15.9	15.3	11.9	8.6	5.0	2.5
	10%	DB	0.3	0.6	4.0	8.4	14.4	19.1	21.8	20.4	15.0	9.3	4.8	2.2
		MCWB	0.0	-0.2	1.1	3.6	8.1	11.8	15.3	15.0	11.8	8.0	4.1	1.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.4	3.7	5.1	9.0	14.0	17.8	20.1	19.7	15.6	11.0	8.1	4.5
		MCDB	5.3	6.0	9.2	15.6	20.6	24.6	25.7	23.9	17.1	11.6	9.3	4.6
	2%	WB	2.1	2.1	3.8	6.9	12.1	15.7	18.6	17.8	14.5	9.9	6.5	3.5
		MCDB	2.9	3.9	7.3	12.0	17.8	19.9	22.8	20.7	16.1	10.8	7.1	3.9
	5%	WB	1.0	0.8	2.6	5.4	10.2	14.5	17.5	16.9	13.5	9.1	5.4	2.5
		MCDB	1.5	1.7	5.3	9.1	15.1	18.1	21.0	19.6	14.8	10.1	5.8	3.0
	10%	WB	0.0	-0.1	1.5	3.9	8.9	13.3	16.6	16.1	12.7	8.0	4.3	1.5
		MCDB	0.3	0.5	3.3	7.4	13.3	16.9	19.9	18.8	14.0	9.3	4.7	2.2
Mean Daily Temperature Range	5% DB	MDBR	5.0	5.8	6.7	7.3	8.1	8.5	8.0	7.5	6.9	5.5	4.1	4.6
		MCDBR	4.8	6.6	8.6	10.5	11.0	11.7	10.6	10.0	8.7	6.4	4.1	4.6
		MCWBR	4.4	5.2	5.4	5.6	5.7	5.5	4.9	4.6	4.6	4.5	3.9	4.3
	5% WB	MCDBR	4.7	5.4	7.9	10.1	10.2	9.1	8.8	8.3	6.6	5.5	4.0	4.2
		MCWBR	4.4	4.7	5.4	5.9	6.1	5.2	4.9	4.6	4.4	4.3	3.9	4.2
Clear-Sky Solar Irradiance	taub	N/A	0.276	0.303	0.336	0.341	0.344	0.360	0.349	0.321	0.303	0.344	0.348	
	taud	N/A	2.164	2.227	2.295	2.464	2.506	2.480	2.507	2.560	2.397	2.611	2.624	
	Ebn at Noon	N/A	624	777	836	870	875	846	823	777	607	316	156	
	Edh at Noon	N/A	59	85	101	94	93	93	82	64	48	23	13	
All-Sky Solar Radiation	RadAvg	0.12	0.68	2.14	3.76	4.97	5.54	5.09	3.84	2.25	0.87	0.20	0.05	
	RadStd	0.01	0.12	0.23	0.35	0.50	0.44	0.47	0.35	0.28	0.16	0.04	0.00	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (47 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page